

JIE HUANG

PERSONAL DATA

GENDER: Male
 NATIONALITY: Chinese
 ADDRESS: Kide 2501, Department of Applied Physics, Aalto University, Finland
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 WEBSITE: <https://way2ml.github.io>

EDUCATION

05/2023–now	Doctoral Researcher in Applied Physics , SIN group Aalto University, Helsinki, Finland, Advisor: Prof. Adam S. Foster
09/2018–06/2021	M.S. in Condensed Matter Physics Wenzhou University, Wenzhou, China, Advisor: Prof. Shibei Li Thesis: “ Applications of machine learning in hydrogen-bond dynamics in bulk water and polymer chains’ structure factor ” GPA: 3.84/4.0
09/2012–06/2016	B.S. in Electronic Information Engineering China West Normal University, Nanchong, China Thesis: “ A CCD translation device based on Raspberry Pi ” GPA: 3.22/4.0

SELECTED PUBLICATIONS

1. Paul Laubrock, Jonas Heggemann, **Jie Huang**, Adam S. Foster*, Philipp Rahe*, [Chiral Water Structures on the Dolomite\(104\)-S Surface Enantiomer](#), **Nano Letters**, DOI: 10.1021/acs.nanolett.6c03227, 2026
2. **Jie Huang**, Niko Oinonen, Fabio Priante, Filippo Federici Canova, Lauri Kurki, Chen Xu, and Adam S. Foster*, [Improving atomic force microscopy structure discovery via style-translation](#), **npj Computational Materials**, DOI: 10.1038/s41524-026-02243-2, 2026
3. Jonas Heggemann, **Jie Huang**, Simon Aeschlimann, Simon Spiller, Adam S. Foster*, and Philipp Rahe*, [Sidestepping Intermolecular Hydrogen Bonds: How Single Water Molecules Adsorb and Assemble on the Calcite \(104\)-\(2x1\) Surface](#), **ACS Nano**, DOI: 10.1021/acsnano.5c05845, 2025
4. Gang Huang and **Jie Huang***, [Revisiting the Thickness of the Air-Water Interface from Two Extremes of Interface Hydrogen Bond Dynamics](#), **Journal of Chemical Theory and Computation**, DOI: 10.1021/acs.jctc.4c00457, 2024
5. **Jie Huang**, Bowen Wang, Hejin Yan, and Yongqing Cai*, [Mechanism of interaction of water above the methylammonium lead iodide perovskite nanocluster: size effect and water-induced defective states](#), **The Journal of Physical Chemistry Letters**, DOI: 10.1021/acs.jpclett.3c02807, 2024
6. **Jie Huang**, Gang Huang*, and Shibei Li*, [A machine learning model to classify dynamic processes in liquid water](#), **ChemPhysChem**, DOI: 10.1002/cphc.202100599, 2022, **Cover Article**
7. **Jie Huang**, Shibei Li*, Xinghua Zhang*, and Gang Huang, [Neural network model for structure factor of polymer systems](#), **The Journal of Chemical Physics**, DOI: 10.1063/5.0022464, 2020

WORK EXPERIENCE

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|-------------------|--|
| 07/2022 - 04/2023 | Research Assistant, Institute of Applied Physics and Materials Engineering, University of Macau , Macau, China. Advisor: Prof. Yongqing Cai
1. Density Functional Theory Study of Perovskite Nanoclusters in Aqueous Environment |
| 09/2016 - 04/2018 | Software Engineer , FalconPro Technology Co., Ltd, Shanghai, China
1. Image processing & Barcode decoding
2. Embedded systems programming |

AWARDS

- | | |
|------|--|
| 2024 | Third Place in AI and ML for Microscopy, Hackathon |
| 2021 | Outstanding Graduate of Wenzhou University |
| 2021 | Award for Excellent Scientific and Technological Innovation |
| 2020 | National Scholarship , Ministry of Education of China |
| 2015 | National Undergraduate Electronic Design Contest (TI Cup) Second Prize |
| 2015 | National Professional Software Engineering Design Contest (Bule Bridge Cup) Second Prize |

CERTIFICATES

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|------|---|
| 2024 | Coursera certificate from DeepLearning.AI : Generative Adversarial Networks (GANs) |
| 2018 | MOOC certificate from Peking University : Artificial Intelligence Practice |
| 2016 | edX certificate from the University of Texas at Austin : Embedded Systems |
| 2015 | National Electronics Professionals : Senior Development Engineer of MCU Application |
| 2014 | National Computer Rank Examination Level 3 : Embedded System Development |

SKILLS

Machine Learning, Molecular dynamics (MD) simulations, Embedded Systems Programming, Linux, Git, Tennis

LANGUAGES

Chinese (native), English (proficient), Japanese (basic), Python, C

ACTIVITIES

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|-------------------|--|
| 01/2025 - 06/2026 | Teaching assistant for PHYS-E0412 - Computational Physics D, Aalto University |
| 2021/11 | Teaching " Introduction to Deep Learning " for postgraduates in Physics , Wenzhou University. |
| 05/2018 - 11/2019 | System manager of the Linux cluster, Beihang University |

HOBBY PROJECTS

- 2025 | [Sphinx-molview](#): A Sphinx extension for interactive 3D molecular visualization
- 2025 | [UP2Git](#): A Linux system tray app for uploading files to GitHub
- 2022 | [Guitar Simulations From Wave Equations: Music Demo](#)
- 2022 | [Huanbu](#): A cross-platform and full functional metronome app based on Python
- 2020 | [Adversarial Inverse Reinforcement Learning Implementation](#) for Mountain Car
- 2019 | [Steady Hand Game Device](#) Based on Raspberry Pi
- 2016 | [Handheld Game Console](#) Based on Texas Instruments TM4C123G

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